Raed Najeeb

+91-8921249258 raeednajeeb@gmail.com
GitHub LinkedIn Portfolio

Summary

B.Tech graduate in Artificial Intelligence and Data Science with hands-on experience building machine learning and LLM systems using Python and PyTorch. Implemented a GPT-style transformer from scratch, developed a local RAG pipeline, experimented with parameter-efficient fine-tuning, and built an evaluation framework for retrieval and abstention metrics. GATE DA qualified; seeking entry-level AI Engineering, Machine Learning, or LLM Evaluation roles.

Technical Skills

Languages: Python
Machine Learning: PyTorch, Hugging Face Transformers, Neural Networks, Classification, Model Training
LLM Engineering: Retrieval-Augmented Generation (RAG), LoRA, Parameter-Efficient Fine-Tuning, Embeddings, Vector Retrieval, Local LLM Inference
Evaluation: Retrieval Evaluation, Recall@K, Precision@K, Mean Reciprocal Rank (MRR), Abstention Evaluation, Benchmarking, Automated Testing
Libraries and Tools: PEFT, Sentence-Transformers, ChromaDB, Ollama, Pydantic, Pytest, Git, GitHub

Selected Projects

nanoGPT From Scratch | PyTorch 2026

- Implemented a character-level GPT-style transformer in PyTorch, including token embeddings, causal self-attention, multi-head attention, feed-forward layers, residual connections, and the training loop.
- Evaluated model performance using training and validation loss, recording approximately 1.77 training loss and 1.90 validation loss.

Local RAG Pipeline | ChromaDB, Sentence-Transformers, Ollama 2026

- Built a document-question-answering pipeline covering document ingestion, overlapping chunking, embedding generation, vector retrieval, and context-grounded answer generation using a locally served Llama 3.1 8B model.
- Implemented retrieval and answer-generation workflows with deterministic inference, context-grounded responses, and refusal behavior when retrieved documents lacked sufficient information.

Emotion Classification with LoRA | Hugging Face PEFT 2026

- Compared LoRA adaptation with full fine-tuning of DistilBERT on the six-class dair-ai/emotion dataset using held-out evaluation.
- LoRA achieved 91.65% accuracy and 0.8771 macro F1 while updating approximately 1.10% of the model's parameters.

EvalLab | LLM and Retrieval Evaluation Framework 2026

- Built a Python evaluation framework with a CLI runner for assessing retrieval performance and abstention behavior using Recall@K, Precision@K, Mean Reciprocal Rank (MRR), and abstention accuracy.
- Implemented structured JSON evaluation reports and automated test coverage using Pytest, with 18 tests passing.

Education

B.Tech, Artificial Intelligence & Data Science Aug 2026
Vimal Jyothi Engineering College, Kannur CGPA: 7.95/10

Achievement

GATE DA Qualified 2025